



NATIONAL UNIVERSITY OF SINGAPORE

EE5434: Microelectronic Processes and Fabrication

Simulation Project

A report submitted to

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EE543 Simulation Project

Part 1: SProcess Flow Analysis

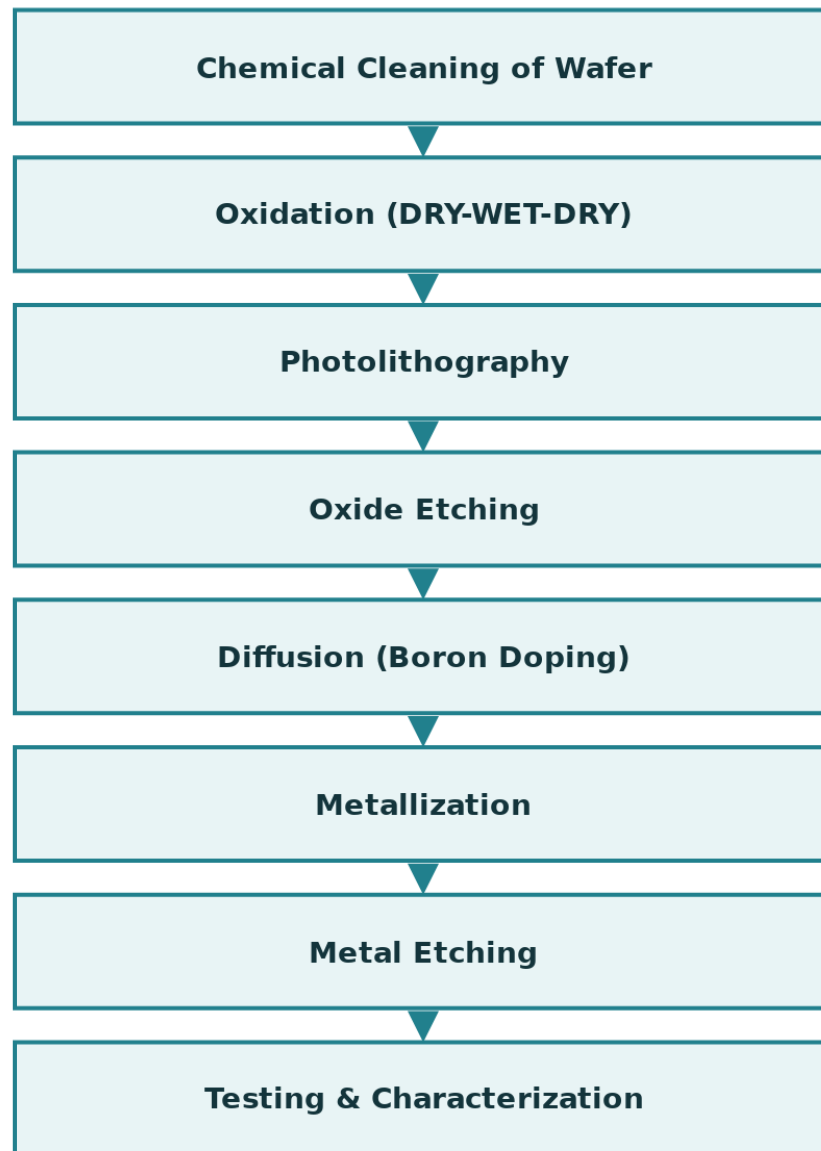


Fig 1: Fabrication Process Flow

Fabrication of PN junction begins by chemical cleaning a silicon wafer to remove impurities. Then a protective oxide layer is then grown, and photolithography patterns the device structure. Next, oxide etching removes unprotected areas to expose the silicon. During diffusion, dopant atoms are introduced into these exposed regions at high temperatures to form the active p-n junction. Afterward, metallization deposits a thin metal layer for electrical contacts, followed by metal etching to isolate the connections. Finally, the completed device undergoes electrical testing and characterization. [1,2]

Initial Oxide Deposition

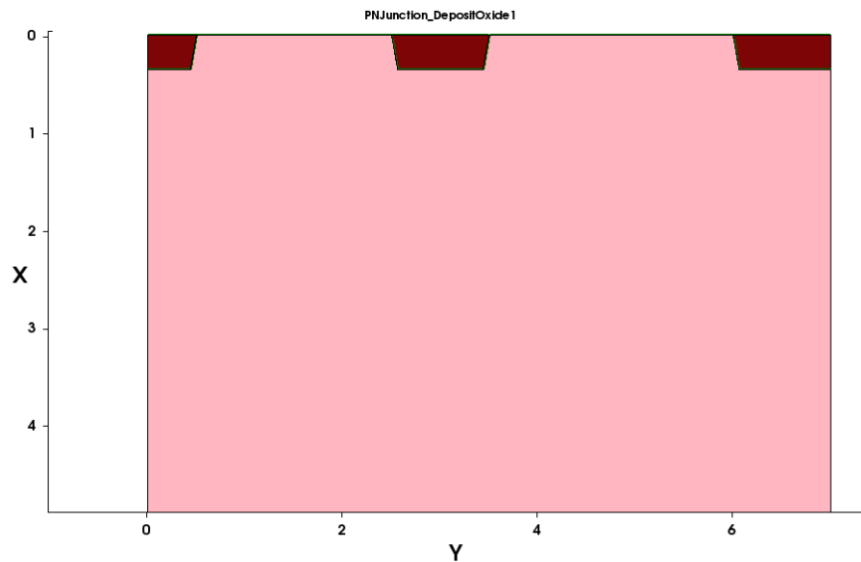


Fig 2: Oxide Deposition

Thermal oxidation or chemical vapor deposition of silicon dioxide. A uniform insulating layer of silicon dioxide is formed across the entire top surface of the bare silicon wafer.

Initial Oxide Etching

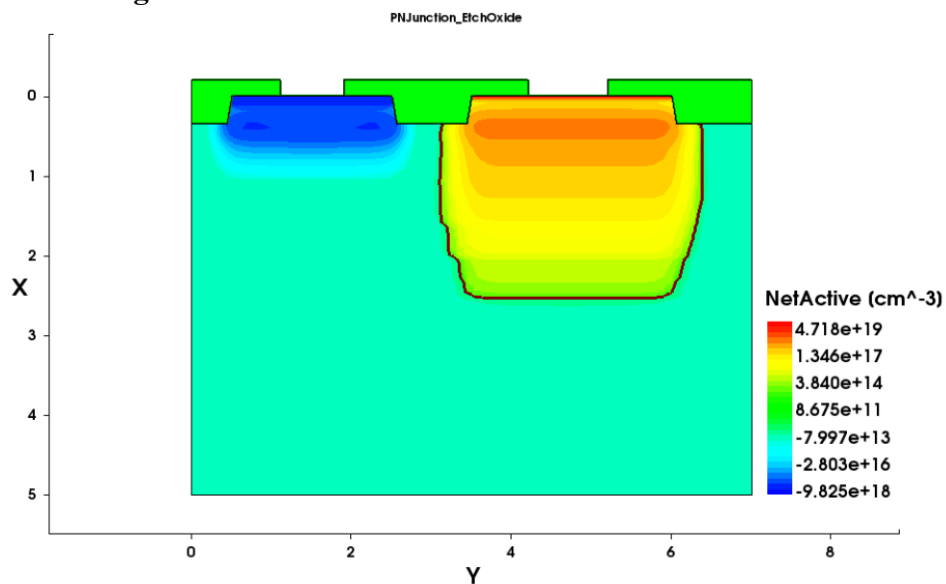


Fig 3: Oxide Etching

A window is etched into this initial oxide layer to expose the underlying silicon and create a hard mask for the dopants. The protective oxide layer is selectively patterned to define the active device area.

Photolithography is followed by selective oxide etching. A targeted window is opened within the continuous oxide layer, exposing the underlying bare silicon substrate.

Ion Implantation

Dopant ions are shot into the exposed silicon window to introduce the necessary charge carriers.

A shallow region of high dopant concentration is created just beneath the surface of the exposed bare silicon. Introducing these specific dopants supplies the majority charge carriers needed to form the N type or P type region.

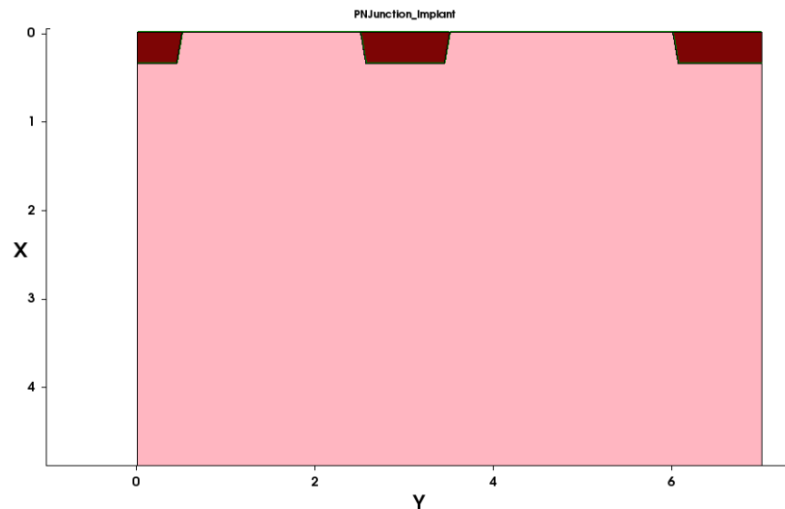


Fig 4: Ion Implantation

Thermal Diffusion

A high temperature treatment is applied to the wafer to redistribute the introduced dopants

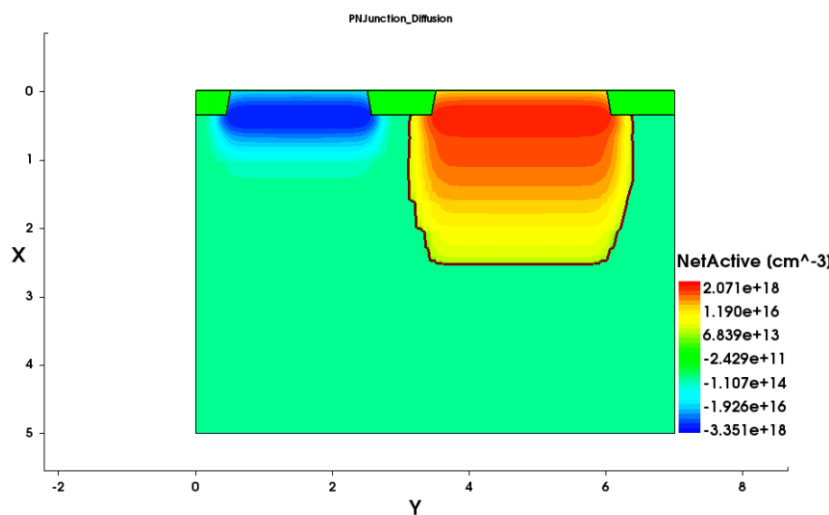


Fig 5: Diffusion

The implanted dopants diffuse deeper into the bulk silicon substrate and spread laterally under the oxide edges, while the severe crystalline lattice damage caused by the energetic implantation process is fully repaired.

Second Oxide Deposition

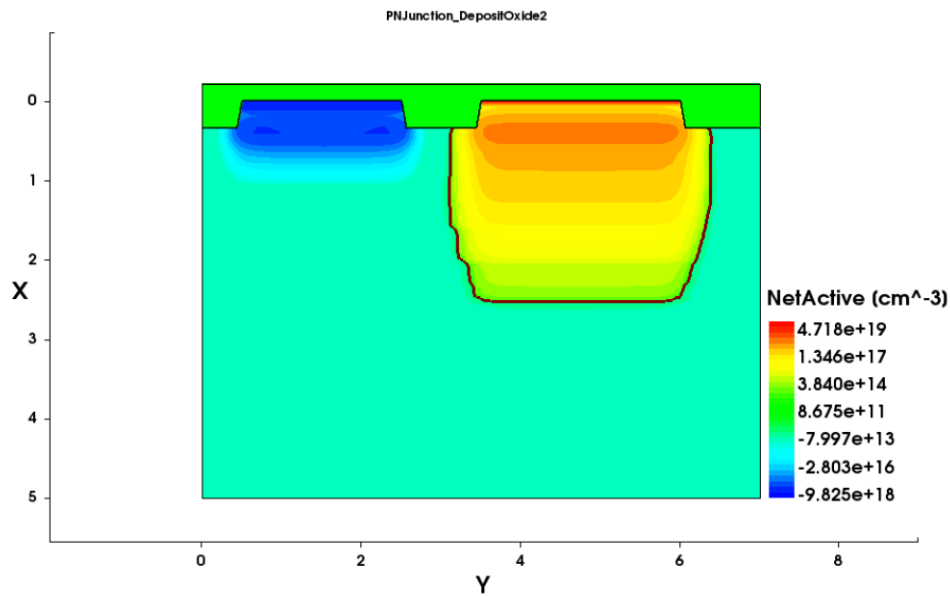


Fig 6: Oxide Deposition

Deposition of a secondary silicon dioxide layer. The previously exposed silicon junction window and the existing field oxide are completely sealed by continuous insulating layer.

Aluminium Deposition

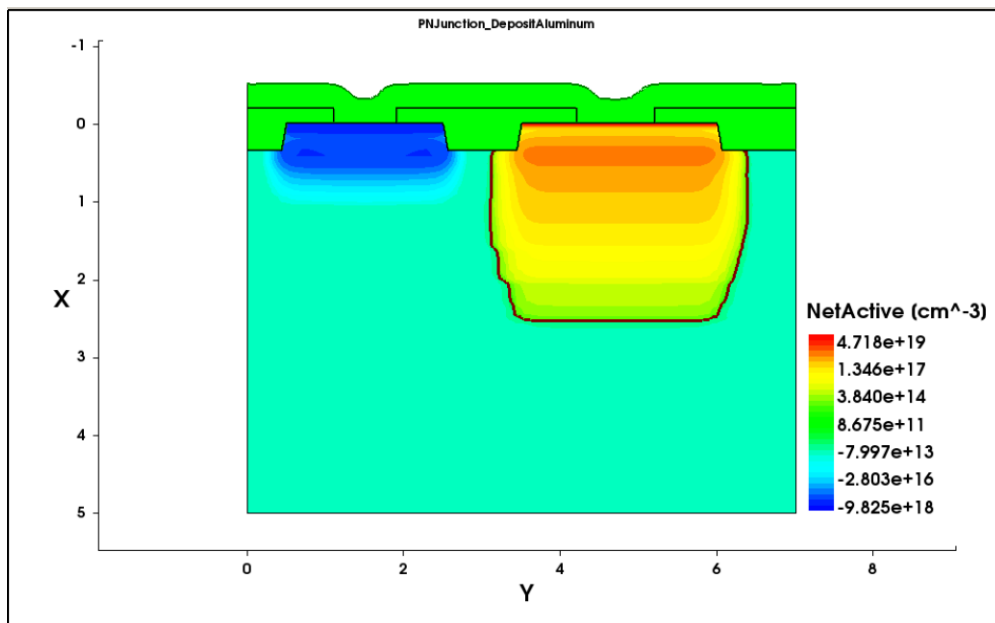


Fig 7: Al Deposition

A blanket layer of highly conductive aluminium is deposited across the wafer, filling the contact hole to touch the silicon. The Metallization could be via physical vapor deposition, thermal evaporation, or sputtering of an aluminium thin film [1,2].

Aluminium Etching

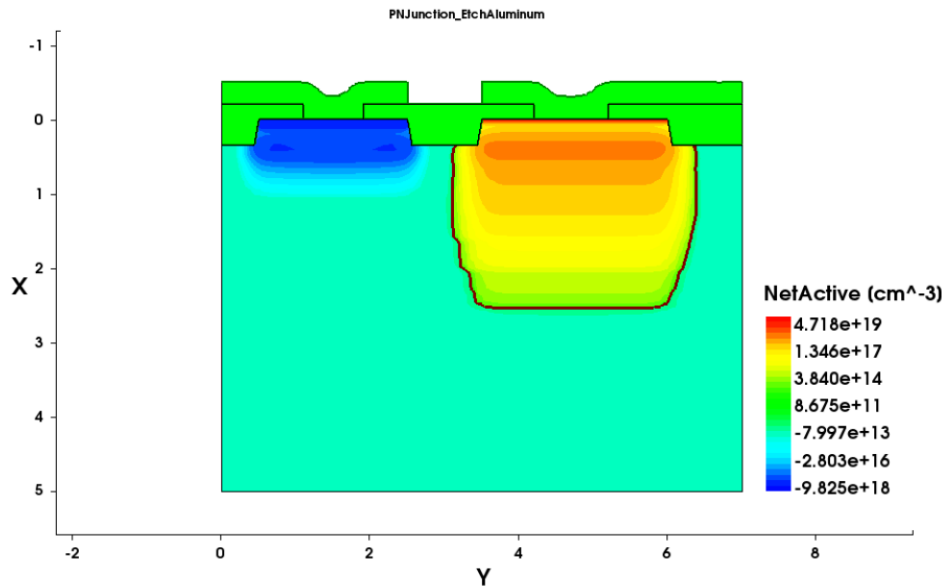


Fig 8: Al Etching

Excess aluminium is stripped away to leave behind a clearly defined metal electrode isolated over the contact opening.

Ohmic contact formation

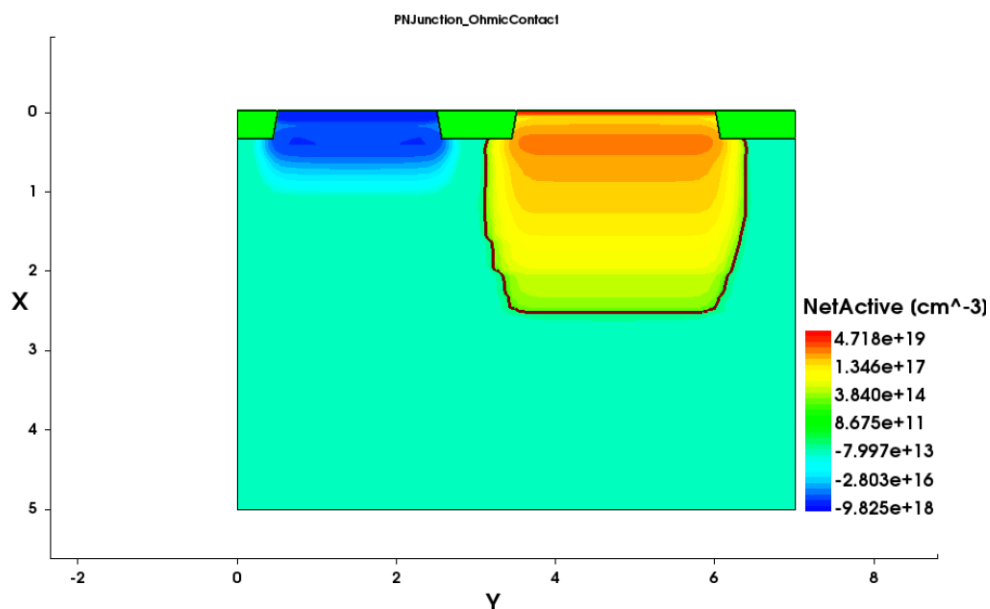


Fig 9: Contact formation

The wafer is annealed so that the patterned aluminium electrode and the heavily doped silicon surface slightly interdiffuse to form a very thin, highly conductive intermetallic alloy layer exactly at the boundary where the metal and silicon touch.

This alloying process is vital because it significantly lowers the interfacial contact resistance and creates a stable non-rectifying (ohmic) electrical connection.

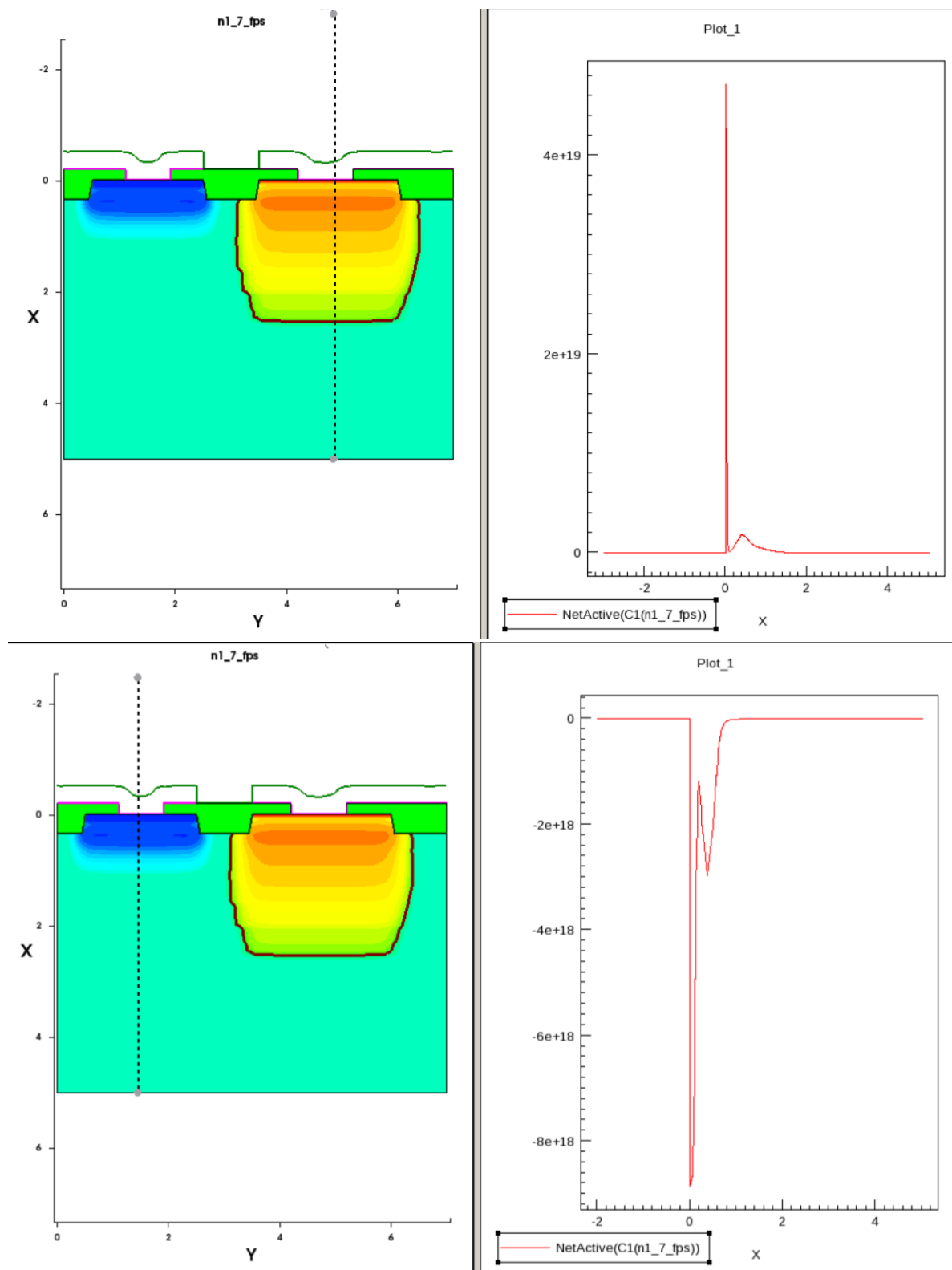


Fig 10: C1 vertical line cross section

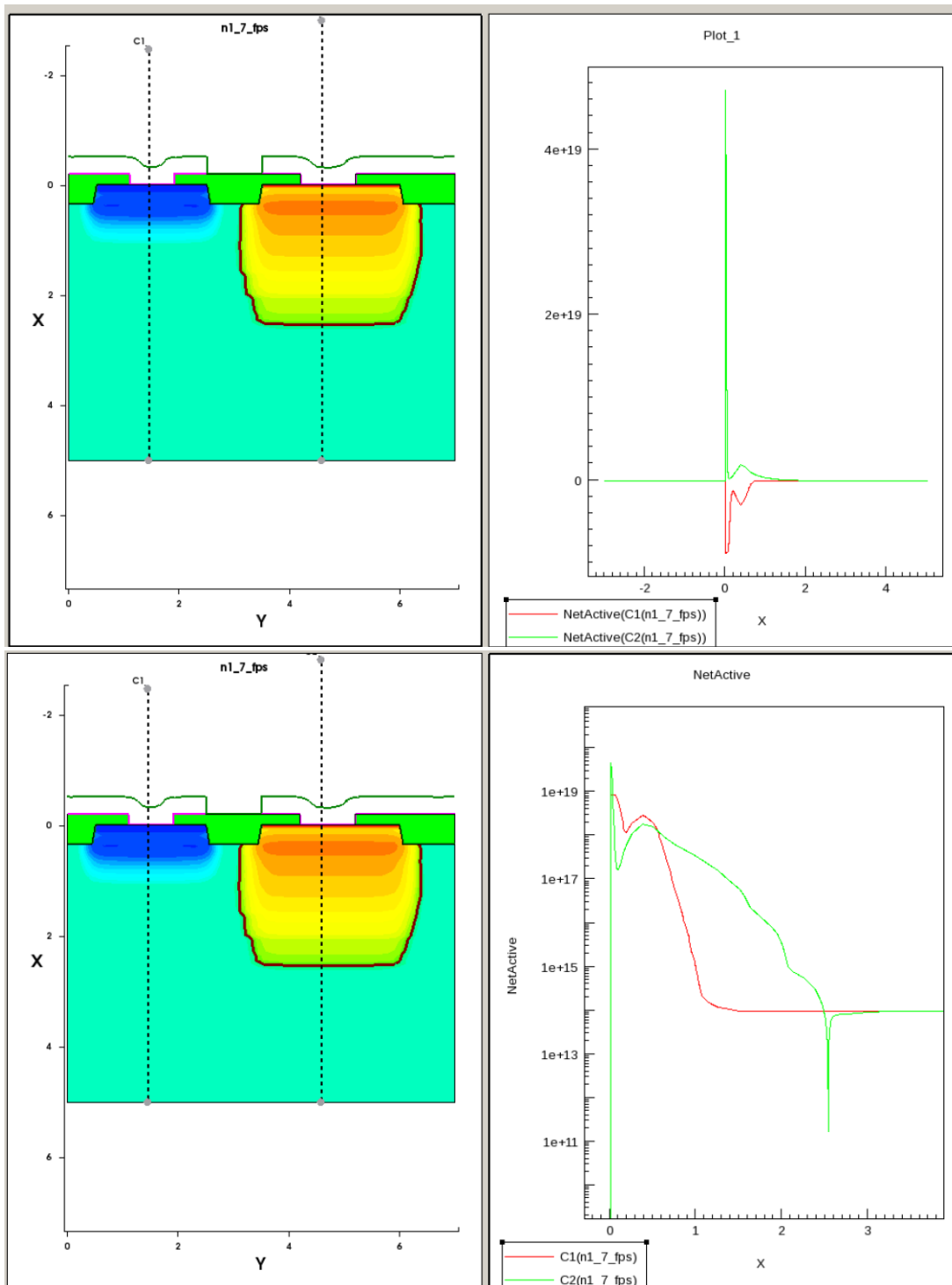


Fig 11: C2 vertical line cross section

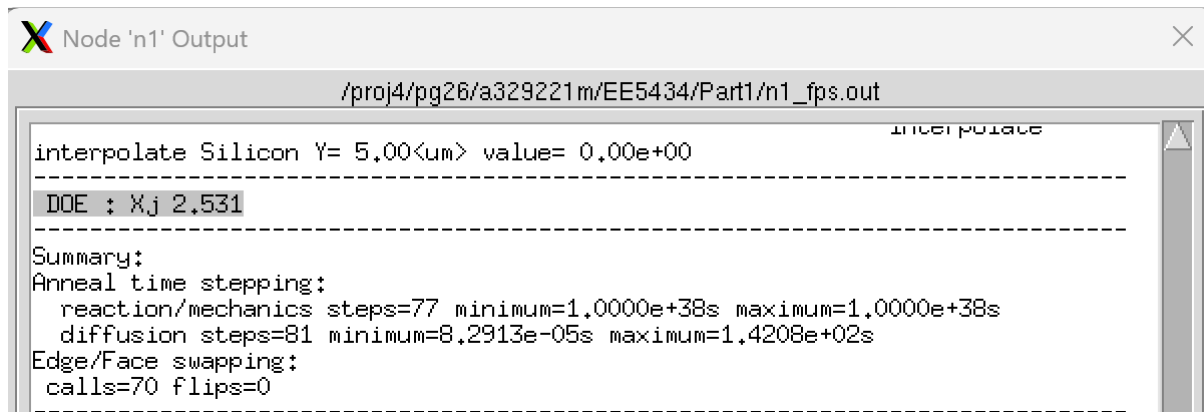


Fig 12: Xj Junction Depth

By drawing vertical cutlines (C1 and C2) through the center of the doped regions, we extract the 1D NetActive Doping Profile to observe carrier concentration versus depth.

From the linear and logarithmic graphs on the right, we can verify the actual metallurgical junction depth, which roughly matches the sharp dip at 2.5 μm on the log scale plot where the doping concentration crosses the background level. Additionally, the logarithmic scale clearly illustrates the massive orders-of-magnitude drop in dopant concentration, confirming the doping gradients created by the thermal diffusion process.

Part 2: Energy Band Diagram Visualization

PN junction device's internal electrostatic variables at zero bias

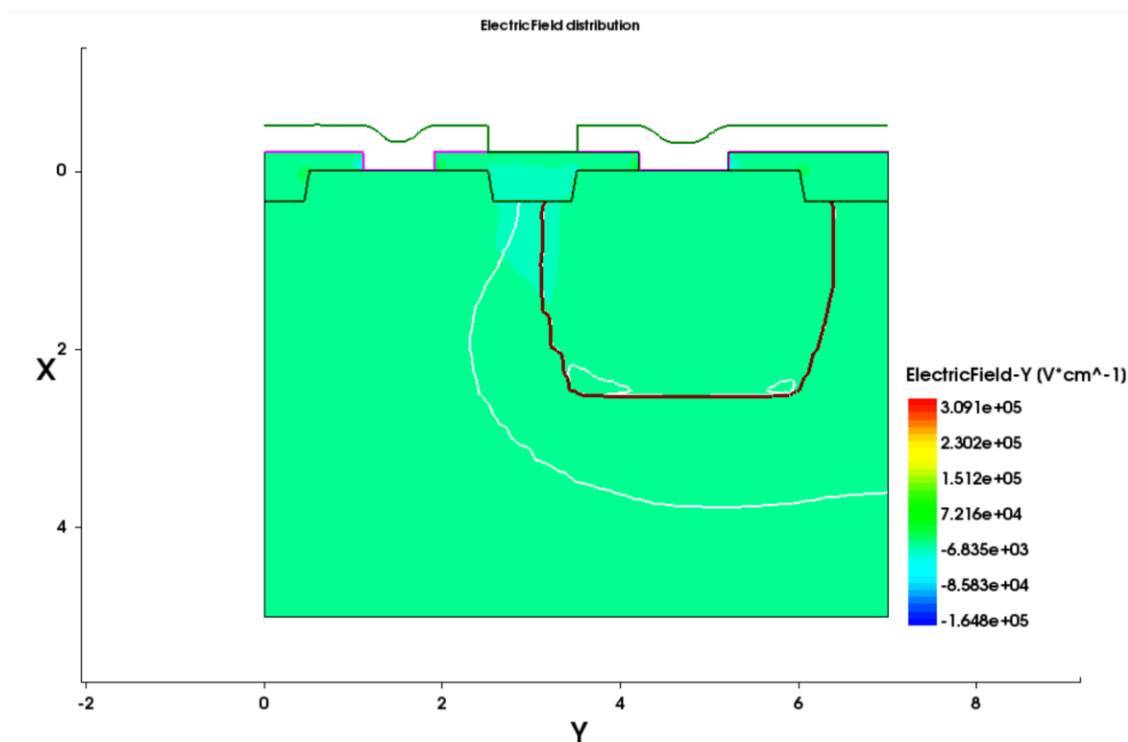


Fig 13: PN Junction Electric Field Distribution

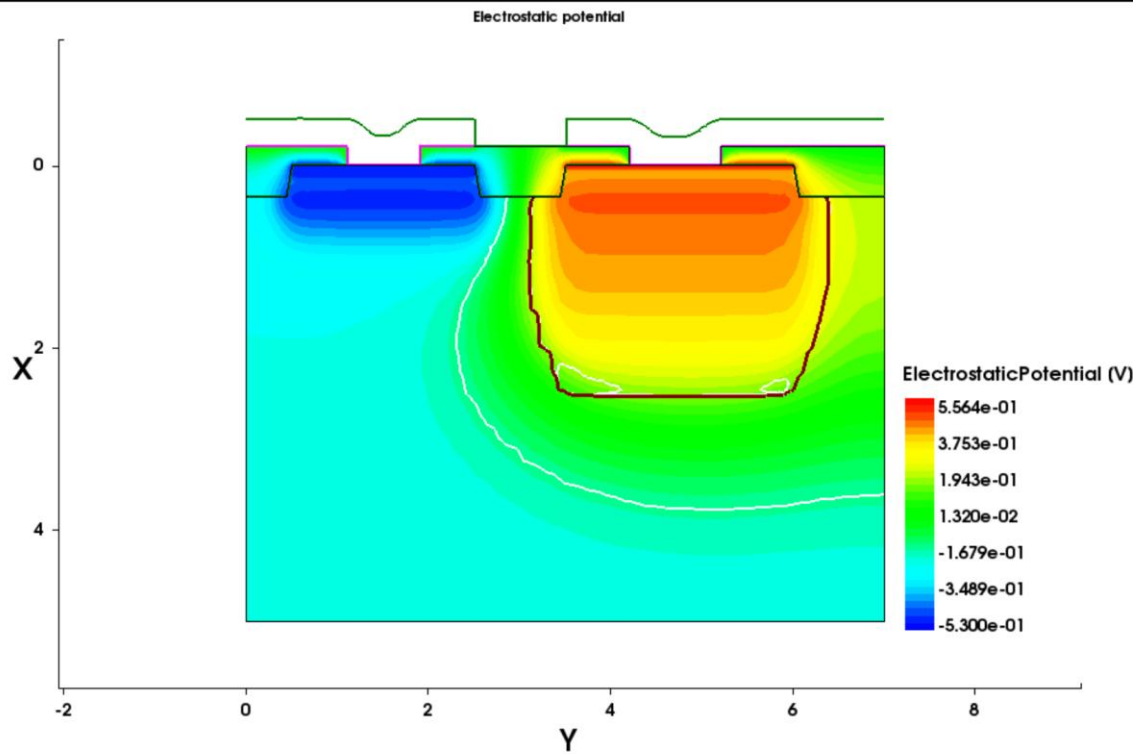


Fig 14: PN Junction Electrostatic Potential

Zero Bias: $V_P = 0V$ (Equilibrium)

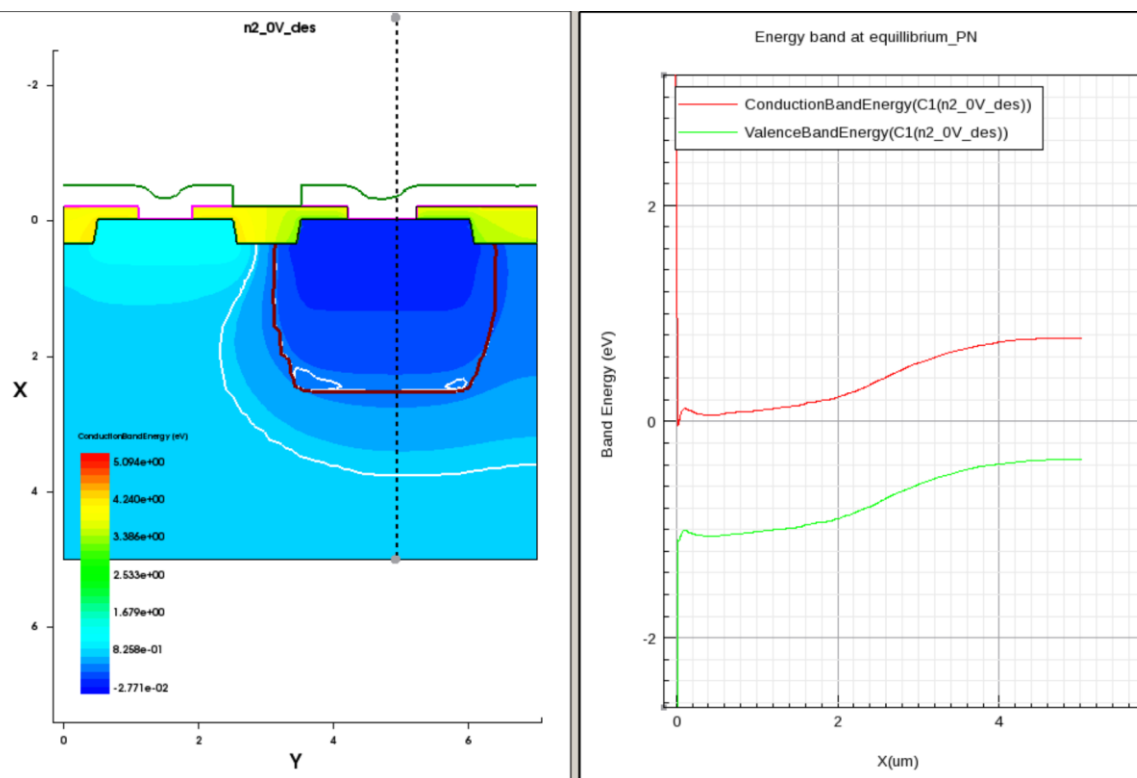


Fig 15: Band Diagram 0 bias

The E_c and E_v smoothly bend upward along the cutline, creating the natural energy barrier of the junction. Because no external voltage is applied, this barrier perfectly balances the carrier flow, keeping the device in equilibrium with zero net current

Forward Bias: $V_P = 0.7V$

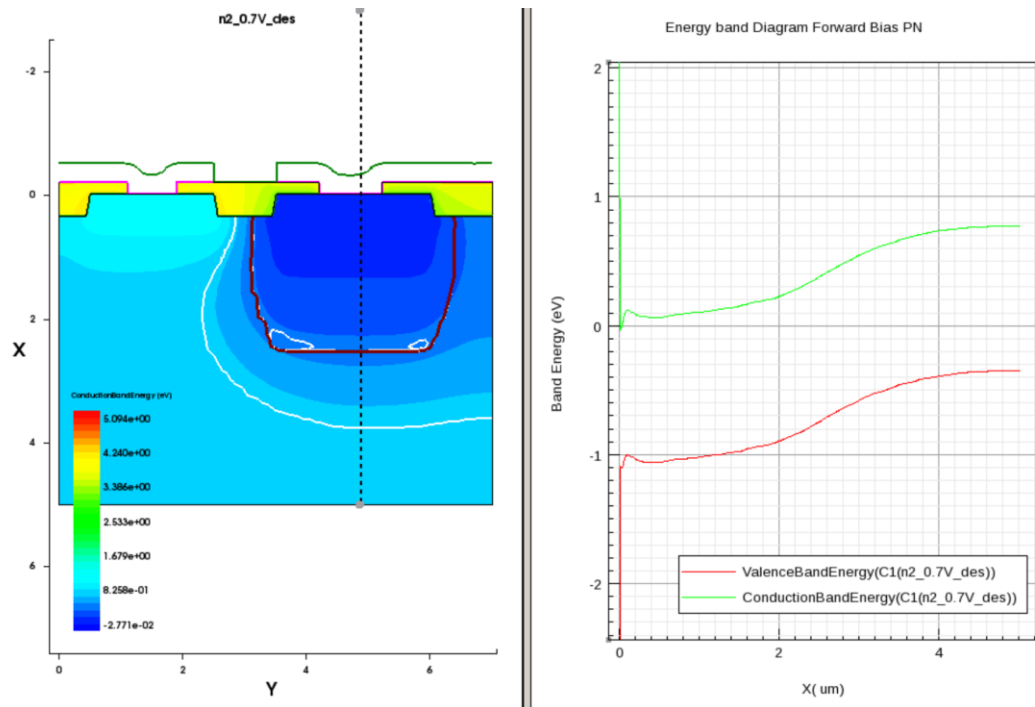


Fig 16: Band Diagram 0.7 V

By applying a 0.7V forward bias, the built-in energy barrier is reduced by 0.7 eV, making the barrier in the conduction and valence bands noticeably flatter than my 0V plot. Because this barrier dropped, it is easy for electrons and holes to cross the junction, which shrinks the depletion region and causes a current (I_d) to flow

Reverse Bias: $V_P = -3V$

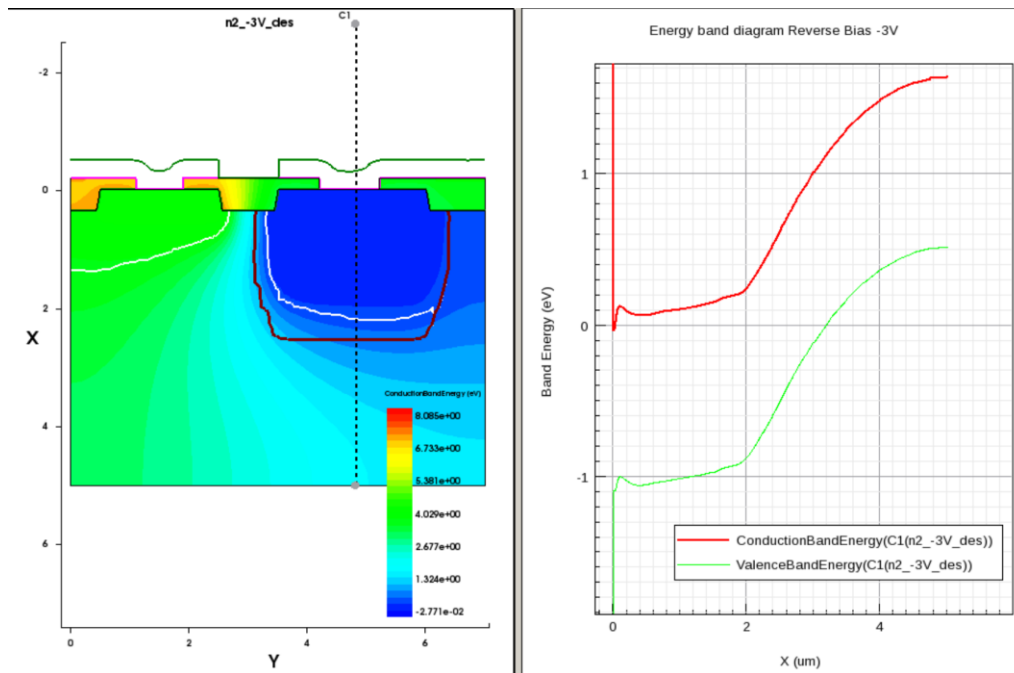


Fig 17: Band Diagram -3V

Applying a -3V reverse bias increases the potential barrier height by exactly 3 eV, creating a potential barrier in the Energy band compared to the equilibrium state. This barrier impacts the majority carriers from crossing, forcing the depletion region to widen significantly (shown by the expanded colour gradients) and restricting the device to just a small current.

To summarize forward bias directly reduces from the built-in potential, shrinking the energy barrier and letting current flow easily. Conversely, reverse bias adds directly to the built-in potential, building a huge energy wall that widens the depletion zone and blocks current.

Part 3: Study of P-type Doping

Observe how the substrate background concentration, also the P type region doping concentration N_A dictates both the physical structure and the electrical behaviour of the diode.

I-V Characteristics: Overlay the I-V curves for all three cases on a single plot for direct comparison.

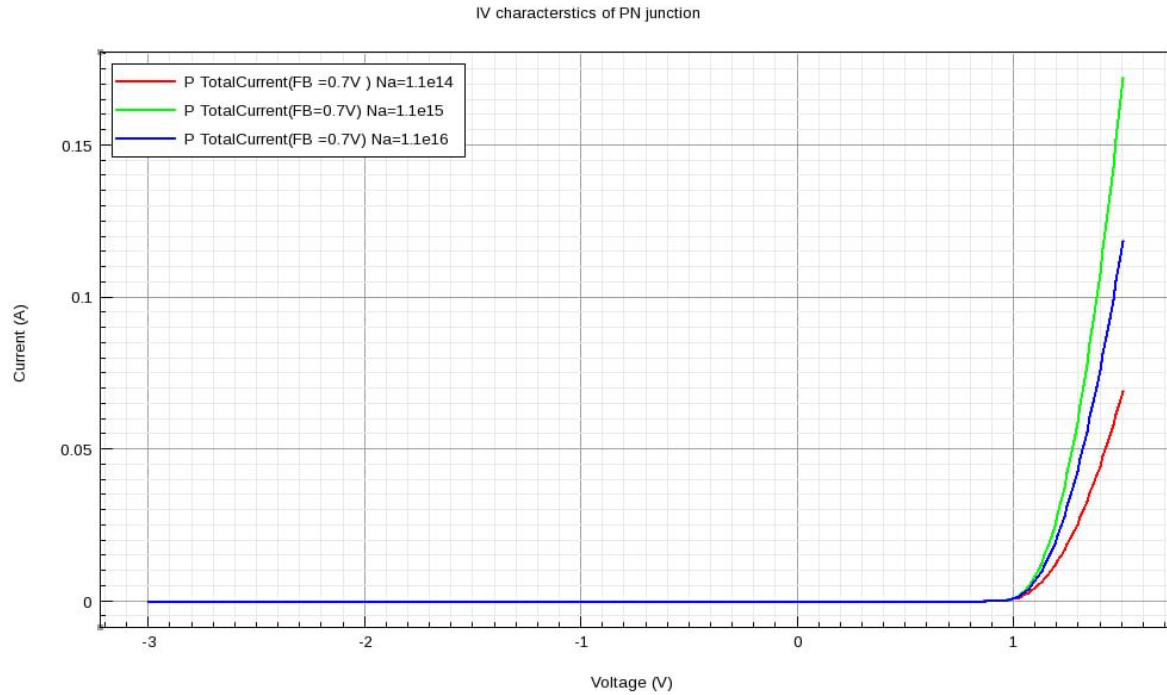


Fig 18: PN Junction IV Characteristics

As the p type substrate doping concentration N_a increases from $1.1 \times 10^{14} \text{ cm}^{-3}$ to $1.1 \times 10^{15} \text{ cm}^{-3}$ and $1.1 \times 10^{16} \text{ cm}^{-3}$, the extracted junction depth decreases from 2.526 to 1.816 because the donor diffusion profile intersects the higher background acceptor concentration closer to the surface.

		 SPROCESS	 SDEVICE	
		Ncon		Xj
1	--	1.1e14	--	2.526
2		11000000000000000.0	--	2.055
3		110000000000000000.0	--	1.816

Fig 19: Sentaurus Test Bench for Ncon and Xj

Increasing N_a and its effect on IV

Reduced Reverse Saturation Current, Increasing the acceptor doping concentration (N_a) reduces the minority carrier concentration on the p-side ($n_{p0} = n_i^2 / N_a$). This leads to a lower leakage current, which shifts the horizontal value of the semi-log plot downward.

$$I = I_0 \left(e^{qV_A/kT} - 1 \right), \quad \text{where } I_0 \equiv qA \left(\frac{D_p n_i^2}{L_p N_D} + \frac{D_n n_i^2}{L_n N_A} \right)$$

Increases turn on voltage, A higher N_a increases the built-in potential (V_{bi}) of the junction. Consequently, the diode requires a higher forward voltage to overcome the potential barrier and reach the same current levels, shifting the exponential region of the curve to the right.

The semilog I V plot shows that higher N_a produces lower reverse current and a slight right shift in the small forward bias region, while at high forward bias all curves merge because the current becomes dominated by strong carrier injection rather than the initial substrate doping variation

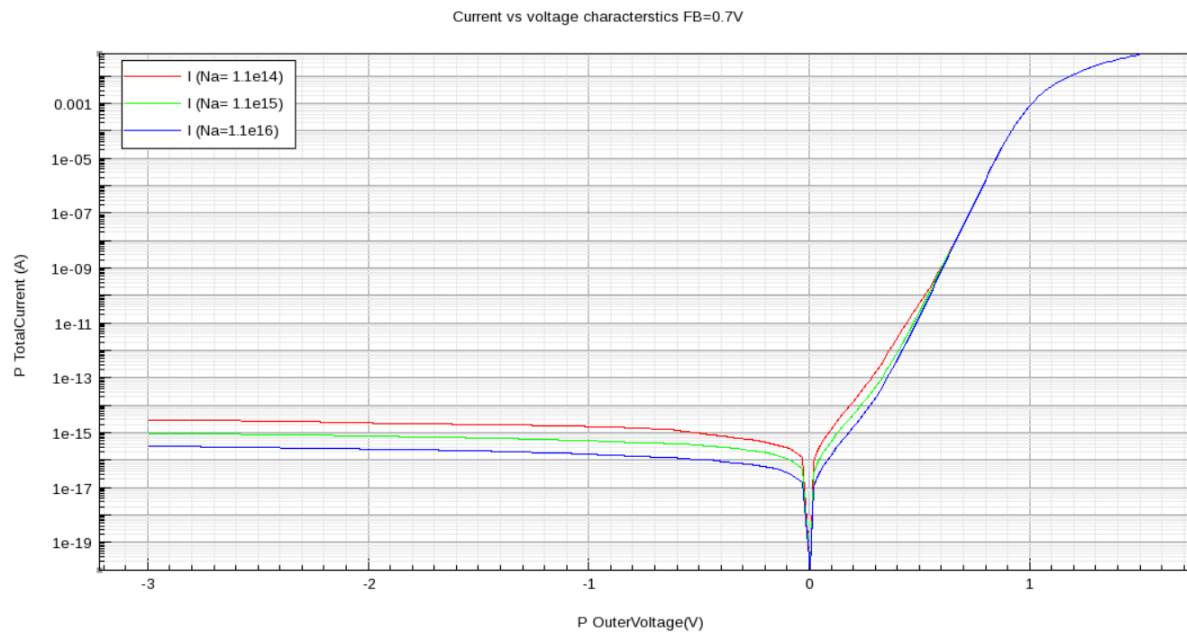


Fig 20: Forward Bias IV Characteristics 0.7V for different N_a

Improves Conductivity, Higher doping reduces the resistivity of the neutral p-type region. This minimizes the ohmic voltage drop (series resistance) at high current levels, allowing the device to maintain its ideal exponential slope for a longer range before the current begins to saturate.

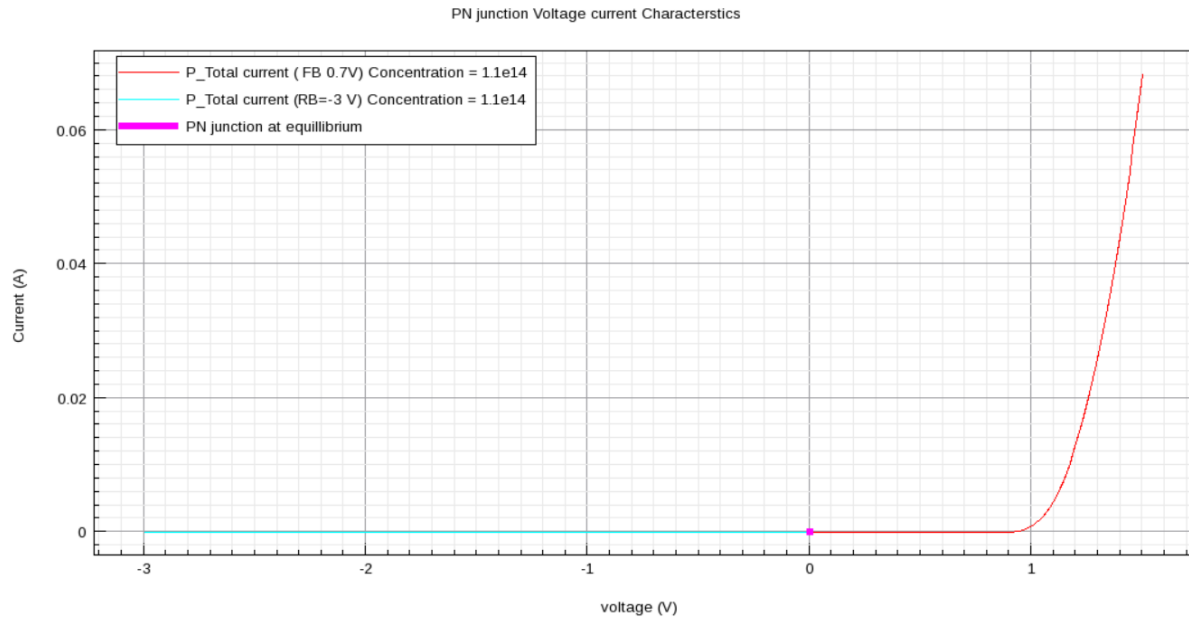


Fig 21: IV Characteristics for 0.7 and -3V

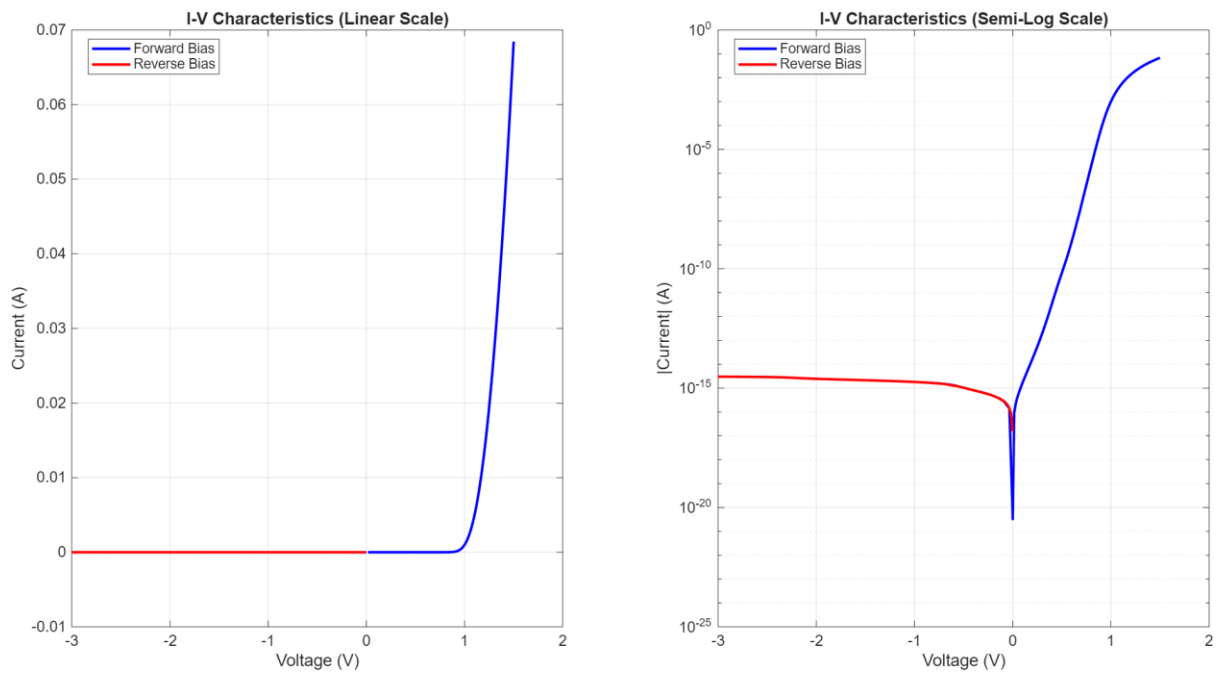


Fig 22: IV Characteristics in Linear and Semi-Log Scale

The behaviour is consistent with established semiconductor theory that the reduction in I_0 and the subsequent right-shift in turn-on voltage are direct consequences of the increased built-in potential and reduced minority carrier concentration associated with higher acceptor doping levels [3].

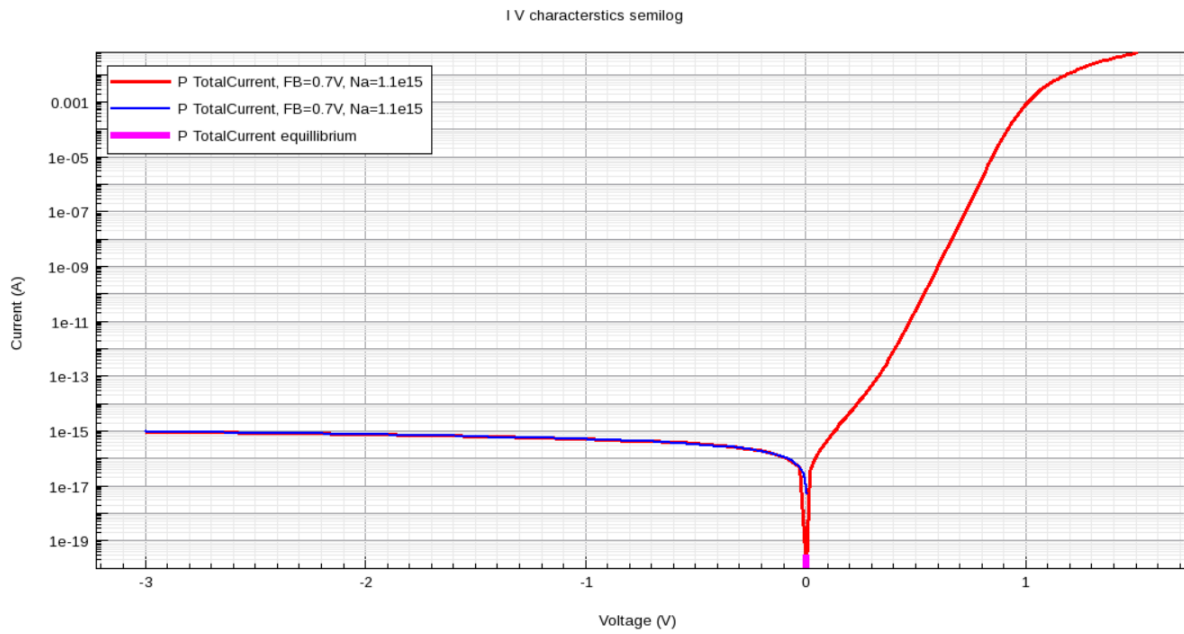


Fig 23: IV Characteristics for Na = 1.1e15

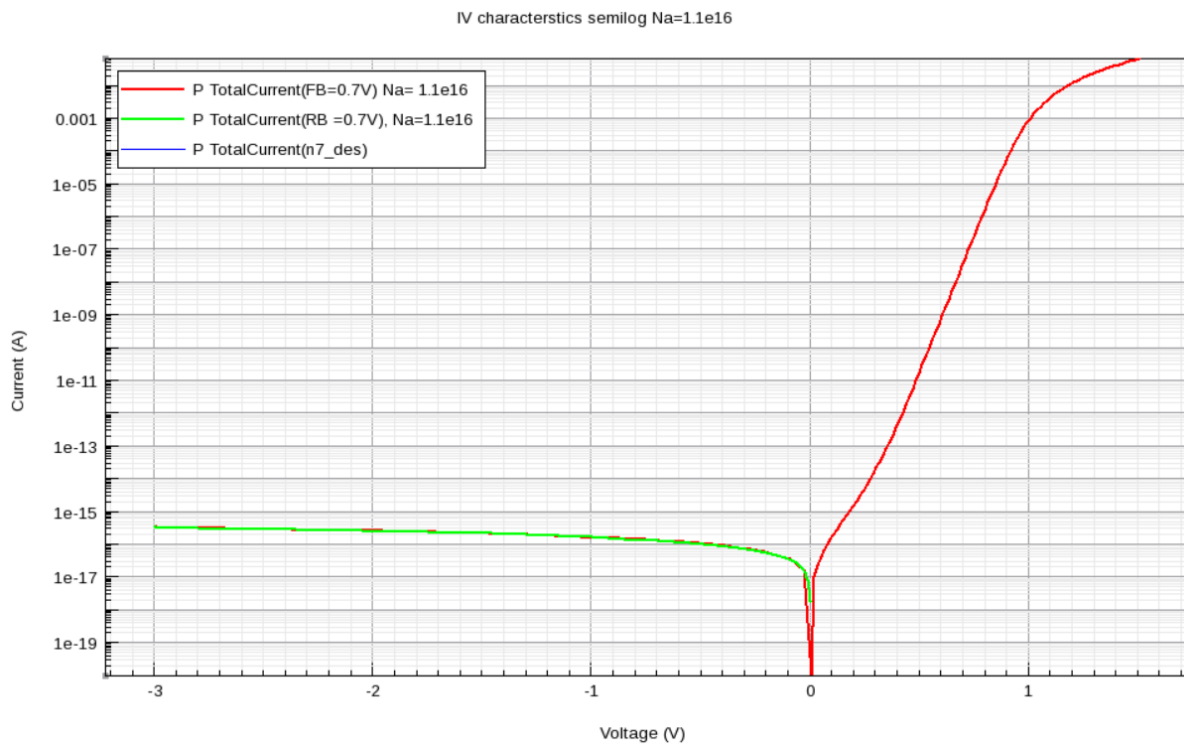


Fig 24: IV Characteristics for Na = 1.1e16

At high current levels, the I-V curves deviate from the ideal slope due to two key phenomena:

High-Level Injection (HLI): When the concentration of injected carriers exceeds the background doping (Na), the ideality factor increases. This "flooding" effect causes the current slope to degrade on the semi-log plot.

Series Resistance (R_s): At the highest bias, the ohmic resistance of the neutral regions and contacts causes a voltage drop ($V = I * R_s$). While higher N_a reduces p-region resistivity slightly delaying this "rollover" all curves eventually converge as these ohmic losses and HLI physics dominate over the initial doping levels.

References

- [1] Smith, Ryan P., et al. "Introduction to semiconductor processing: Fabrication and characterization of pn junction silicon solar cells." *American Journal of Physics* 86.10 (2018): 740-746.
- [2] P. Choudhary and M. Choudhary, "Fabrication of p-n Junction Diode Using Device Fabrication Technology," *SKIT Research Journal*, vol. 7, no. 2, pp. 25-28, 2017.
- [3] Physics of Semiconductor Devices Author(s):S. M. Sze, Kwok K. Ng

Appendix

SPROCESS CODE:

```
#-----
# (1) Basic Settings: Define Substrate and Simulation Domain
#-----
set Tsi 5.0
set Ymax 7.0
set Ymin 0.0
set Tmetal 0.3

# Pre-calculate spacing variables for compatibility
set Tsi_spacing [expr $Tsi / 5.0]

math coord.ucs
math numThreads= 4

# Define Mesh Lines
line x location= 0.0<um> spacing= 0.01<um> tag= top
line x location= $Tsi<um> spacing= $Tsi_spacing<um> tag= bottom
line y location= $Ymin<um> spacing= 0.1<um> tag= left
line y location= $Ymax<um> spacing= 0.1<um> tag= right

# Define Simulation Region
region Silicon substrate xlo= top xhi= bottom ylo= left yhi= right

# Initialize P-type substrate with Boron concentration of 1E14
init concentration= @Ncon@ field= Boron wafer.orient= { 0 0 1 } \
  flat.orient= { 1 1 0 } slice.angle= -90<degree>

# Mesh Control:
pdbSet Grid Adaptive 1
pdbSet Diffuse Diffuse.Regrid.Steps 5

grid remesh
mgoals accuracy= 1e-5<um> resolution= 0.25

# Initial Adaptive Mesh Refinement
refinebox Adaptive name= AdpMesh \
  refine.fields= { NetActive } \
  def.max.asinhdiff= 0.8 \
  refine.min.edge= " 0.03 0.03 " \
  refine.max.edge= " 1 1 " \
  materials= {Silicon}
grid remesh
```

```

#-----
# (2) Define Active Region: Shallow Trench Isolation (STI)
#-----
mask name= ACTIVE segments= { 0.5 2.5 3.5 6.0 } negative
photo mask= ACTIVE thickness= 1.0<um>
etch material= { Silicon } type= trapezoidal rate= 0.35<um/min> time= 1.0<min> angle= 80<degree>
strip Resist
struct tdr= "n@node@_EtchSilicon"

deposit material= Oxide type= fill coord= 0.0
struct tdr= "n@node@_DepositOxide1"
#-----
# (3) Form NWELL and PWELL: Core PN Junction Formation
#-----
mask name= NWELL segments= {3.5 6}
photo mask= NWELL thickness= 1.0<um>
implant Phosphorus dose= 1E14 energy= 250<keV> tilt= 0
strip Resist

mask name= PWELL segments= {0.5 2.5}
photo mask= PWELL thickness= 1.0<um>
implant Boron dose= 1E14 energy= 100<keV> tilt= 0
strip Resist

# Snapshot: Post-Well Implantation (As-implanted)
struct tdr= "n@node@_Implant"

#-----
# (4) Anneal: Drive-in Diffusion
#-----
diffuse temperature= 900<C> time= 10<min>

# Snapshot: Post-Diffusion (Junction Drive-in)
struct tdr= "n@node@_Diffusion"

# Extract the junction depth ( Xj ) along a vertical cut at Y =5.0 um
# The metallurgical junction is defined as the coordinate where NetActive = 0

select z= "NetActive"
set Xj [interpolate Silicon y= 5.0 value= 0.0]
puts "DOE: Xj [format %.3f $Xj]"
#-----
# (5) Form P+ and N+: Ohmic Contacts and Source/Drain
#-----
mask name= Pplus segments= {0.5 2.5}
photo mask= Pplus thickness= 1.0
implant Boron dose= 1e14<cm-2> energy= 5<keV> tilt= 0
strip Resist

```

```

mask name= Nplus segments= { 3.5 6 }
photo mask= Nplus thickness= 1.0
implant Arsenic dose= 1e14<cm-2> energy= 5<keV> tilt= 0
strip Resist

# Final Source/Drain Activation Anneal
diffuse temperature= 1000<C> time= 1.0<min>

# Snapshot: Complete Doping Profile
struct tdr= "n@node@_OhmicContact"

#-----
# (6) Metalization: Contact Formation and Electrode Definition
#-----
deposit material= Oxide thickness= 0.2<um> type= isotropic
struct tdr= "n@node@_DepositOxide2"

mask name= CNT segments= { 1.1 1.9 4.2 5.2 }
photo mask= CNT thickness= 1.0
etch material= Oxide thickness= 0.21<um> type= anisotropic
strip Resist
struct tdr= "n@node@_EtchOxide"

deposit material= Aluminum thickness= $Tmetal<um> type= isotropic
struct tdr= "n@node@_DepositAluminum"

mask name= METAL1 segments= { 2.5 3.5 } negative
etch mask= METAL1 material= Aluminum thickness= [expr $Tmetal * 2.0]<um> type= anisotropic
struct tdr= "n@node@_EtchAluminum"

# Define Logical Electrodes for SDevice
contact name= N point x= -0.2<um> y= 4.5<um> replace
contact name= P point x= -0.2<um> y= 1.5<um> replace

#-----
# (7) REMESH: Grid Refinement for SDevice Electrical Simulation
#-----
line clear
refinebox clear

# Solver-specific mesh settings
pdbSet Grid SnMesh DelaunayType boxmethod
pdbSet Grid sMesh 1
pdbSet Grid SnMesh DelaunayTolerance 0.0

# Mesh refinement based on NetActive concentration gradient
refinebox Adaptive name= Remesh \

```

```

refine.fields= { NetActive } \
def.max.asinhdiff= 0.8 \
refine.min.edge= " 0.01 0.01 " \
refine.max.edge= " 0.4 0.4 " \
materials= {Silicon}

# Interface refinement for Silicon/Oxide boundaries
refinebox interface.mat.pairs= {Silicon Oxide} \
    min.normal.size= 0.01 normal.growth.ratio= 1.5

grid remesh

# Final Snapshot for SDevice Input
struct tdr= "n@node@_7"

```

SDEVICE CODE:

```

#=====
# SDevice Command File: PN Junction Electrical Characterization
#=====

File {
    # Input: Final mesh and doping profile from SProcess
    Grid  = "n@node|sprocess@_7_fps.tdr"

    # Output: Logs, TDR files for visualization, and PLT files for IV curves
    Output = "@log@"
    Plot   = "@tdrdat@"
    Current = "@plot@"
}

Electrode {
    # Names must strictly match the 'contact' labels in the SProcess script
    { Name="P" Voltage=0.0 }
    { Name="N" Voltage=0.0 }
}

Physics {
    # Use Fermi-Dirac statistics (important for highly doped regions)
    Fermi
}

Physics (Material="Silicon") {
    # Recombination models define leakage and carrier lifetime physics
    Recombination (
        SRH    # Shockley-Read-Hall: Thermal recombination via lattice defects
    )
}

```

```

Math {
    # 'Super' is a direct solver, generally more robust for 2D diode simulations
    Method = Super
    Number_of_Threads = 4

    # Prevents wasted CPU time if the simulation diverges
    ExitOnFailure
}

Plot {
    # Variables for visual analysis in SVisual
    Potential SpaceCharge Doping
    eDensity hDensity
    Current/Vector eCurrent/Vector hCurrent/Vector
    ElectricField/Vector
    eQuasiFermi hQuasiFermi
    SRHRecombination
    BandGap BandGapNarrowing
    ConductionBandEnergy ValenceBandEnergy
}

Solve {
    #-----
    # Step 1: Equilibrium Solution
    #-----
    # Solve Poisson only first, then coupled with carriers at 0V
    Poisson
    Coupled { Poisson Electron Hole }
    #Plot(FilePrefix="n@node@_0V")

    #-----
    # Step 2: Reverse Bias Sweep (Characterizing Leakage/Breakdown)
    #-----
    # Note: NewCurrentPrefix separates reverse data from forward data in the .plt file
    NewCurrentPrefix = "Reverse_"
    Quasistationary (
        InitialStep=1e-3 MaxStep=0.05 MinStep=1e-6
        Increment=1.3 Decrement=2.0
        Goal { Name="P" Voltage=-3.0 }
    ) { Coupled { Poisson Electron Hole } }

    #-----
    # Step 3: Forward Bias Sweep (Characterizing Diode Turn-on)
    #-----
    NewCurrentPrefix = "Forward_"

    # Returning to 0V before sweeping forward to ensure convergence

```

```

Quasistationary (
  InitialStep=1e-3 MaxStep=0.02
  Goal { Name="P" Voltage=0.0 }
) { Coupled { Poisson Electron Hole } }
Plot(FilePrefix="n@node@_0V")

Quasistationary (
  InitialStep=1e-3 MaxStep=0.05 MinStep=1e-7
  Increment=1.3 Decrement=2.0
  Goal { Name="P" Voltage=1.5 }
) {
  Coupled { Poisson Electron Hole }
  # Increase Intervals for a smoother IV curve in the plot
  CurrentPlot( Time=(Range=(0.0 1.0) Intervals=100) )
}
}

```